

Week 8: Progress Check

We propose a novel microneedle-embedded drug-eluting stent to deliver matrix metalloproteinase-9 (MMP-9) inhibitors directly into the lipid core of advanced atherosclerotic plaques. By penetrating the fibrous cap using microneedles and targeting macrophage activity beneath, our design aims to stabilize vulnerable plaques and prevent rupture. The microneedles may be fabricated from biocompatible polymers and optionally integrate drug reservoirs (rectangular prisms) for controlled, localized release.

MMP-9 plays a central role in destabilizing atherosclerotic plaques by degrading extracellular matrix components, particularly type IV and V collagen within the fibrous cap. Elevated MMP-9 levels correlate with larger necrotic core size, plaque instability, and adverse cardiovascular outcomes in both coronary artery disease and acute coronary syndromes¹. MMP-9 is not only a biomarker but also a therapeutic target. Its inhibition reduces inflammation, neovascularization, and the risk of plaque rupture.

The fibrous cap of advanced atherosclerotic plaques is typically 65–150 μm thick². Therefore, microneedles approximately 200 μm in height are ideal—they penetrate just past the fibrous cap to reach the lipid core and associated macrophages, without risking full transmural vessel damage. Microneedle systems have shown superior delivery efficiency and mechanical safety in vascular models, such as the microneedle drug-eluting balloon used successfully in vivo in rabbit aortas³. The small size ensures localized, minimally invasive drug transfer even in complex vascular geometries.

JNJ0966 is a first-in-class, allosteric inhibitor that selectively prevents zymogen (proMMP-9) activation without interfering with the catalytic domains of other MMPs⁴. Unlike broad-spectrum inhibitors that failed due to musculoskeletal toxicity, JNJ0966 achieves high specificity by binding to a regulatory site near Arg-106, avoiding disruption of zinc-coordinated catalytic centers. It reduces disease severity in vivo (e.g., in mouse models of neuroinflammation) and is a strong candidate for plaque-stabilizing therapy.

Morin is a naturally derived flavonoid with potent MMP-9 inhibitory, antioxidant, and anti-inflammatory effects. It directly binds MMP-9 and downregulates Notch-1 signaling, which is implicated in endothelial-to-mesenchymal transition, a process contributing to fibrous cap weakening⁵. In both cell culture and ApoE^{-/-} mouse models, morin reduced intimal hyperplasia and plaque formation. It represents a biocompatible, phytotherapeutic alternative or adjunct to synthetic inhibitors like JNJ0966.

Figures

Figure 1: Basic Stent Design

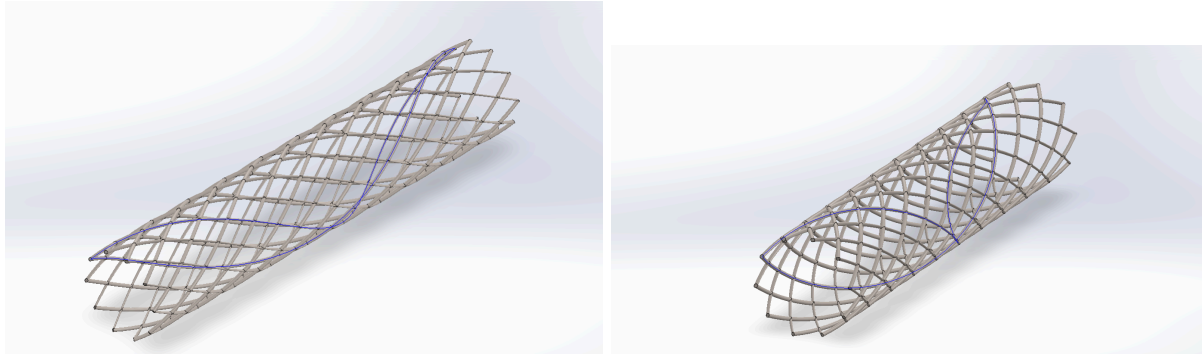
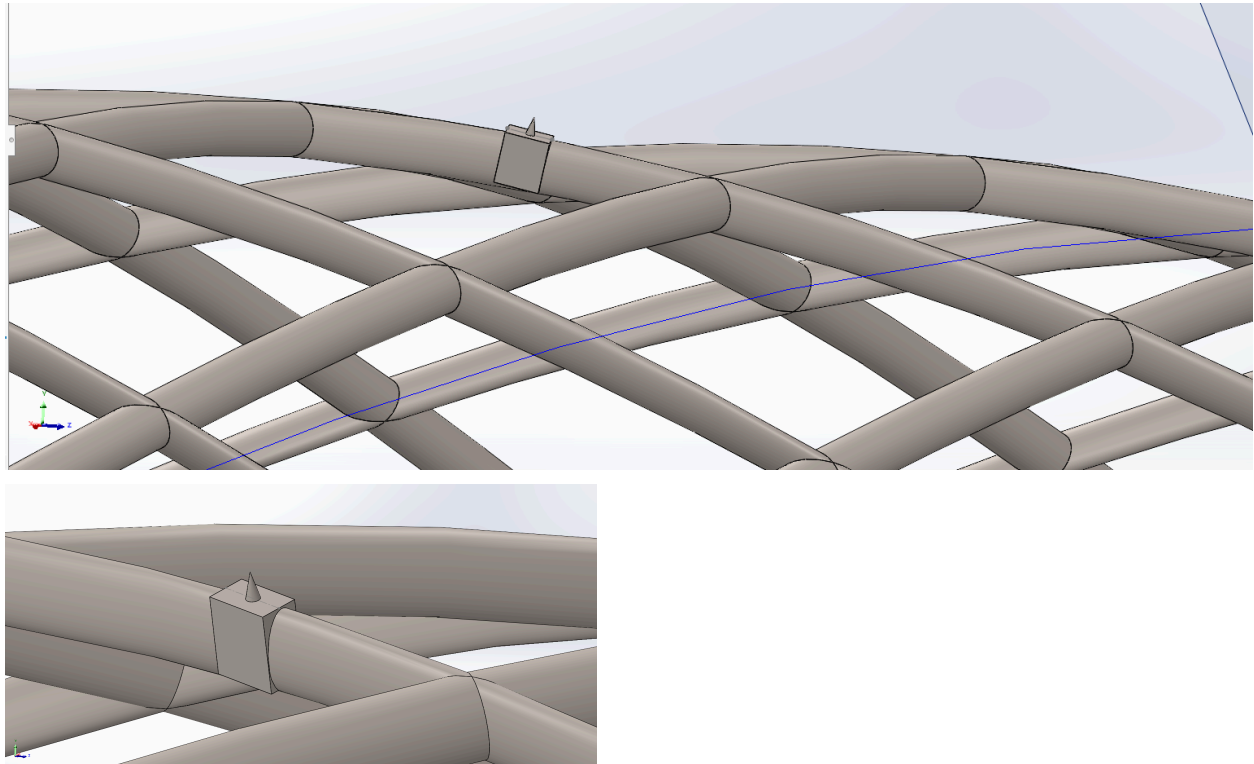


Figure 2: Stent with Microneedle (Cone) and Drug Reservoir (Rectangular Prism)



References

1. Olejarz, W., Łacheta, D., & Kubiak-Tomaszewska, G. (2020). Matrix Metalloproteinases as Biomarkers of Atherosclerotic Plaque Instability. *International journal of molecular sciences*, 21(11), 3946. <https://doi.org/10.3390/ijms21113946>
2. Fayad, Z. A., & Fuster, V. (2001). Clinical imaging of the high-risk or vulnerable atherosclerotic plaque. *Circulation research*, 89(4), 305–316. <https://doi.org/10.1161/hh1601.095596>
3. Lee, K., Lee, J., Lee, S. G., Park, S., Yang, D. S., Lee, J.-J., Khademhosseini, A., Kim, J. S., & Ryu, W. (2020). Microneedle drug eluting balloon for enhanced drug delivery to vascular tissue. *Journal of Controlled Release*, 321, 174–183. <https://doi.org/10.1016/j.jconrel.2020.02.012>
4. Scannevin, R. H., Alexander, R., Haarlander, T. M., Burke, S. L., Singer, M., Huo, C., Zhang, Y. M., Maguire, D., Spurlino, J., Deckman, I., Carroll, K. I., Lewandowski, F., Devine, E., Dzordzorme, K., Tounge, B., Milligan, C., Bayoumy, S., Williams, R., Schalk-Hihi, C., Leonard, K., ... Rhodes, K. J. (2017). Discovery of a highly selective chemical inhibitor of matrix metalloproteinase-9 (MMP-9) that allosterically inhibits zymogen activation. *The Journal of biological chemistry*, 292(43), 17963–17974. <https://doi.org/10.1074/jbc.M117.806075>
5. He, Y., Qin, X. X., Liu, M. W., & Sun, W. (2024). Morin, a matrix metalloproteinase 9 inhibitor, attenuates endothelial-to-mesenchymal transition in atherosclerosis by downregulating Notch-1 signaling. *Journal of integrative medicine*, 22(6), 683–695. <https://doi.org/10.1016/j.joim.2024.11.002>